Faculty Profile
Ram Lal Anand
College, University of

Titl e	Dr.	Name:	Swagata Karmakar	
Address		Department of Environmental Studies, Ram Lal Anand College, 5, Benito Juarez Road, New Delhi-110021, India		
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Google Scholar ID	https://scholar.google.com/citations?user=HsjqggwAAAAJ&hl=en
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Educational Qualifications

Degree	Institution
Ph.D	Environmental Studies, University of Delhi

Permanent/ Temporary/ Contractual/ Adhoc Teaching Experience

Designation	Institution	Nature
Assistant Professor	Ram Lal Anand College, University of Delhi	Permanent
Assistant Professor	Maitreyi College, University of Delhi	Guest Faculty

Assistant Professor	Miranda House, University of Delhi	Guest Faculty
Assistant Professor	School of Open Learning, University of Delhi	Guest Faculty
Assistant Professor	Keshav Mahavidyalaya, University of Delhi	Guest Faculty

Areas of Interest / Specialisation
Soil Microbial Ecology, Microbial Interactions, Remediation, phage therapy, Bioprospecting
Subjects Taught
Environmental Science, Ecosystem, Natural Resources, Biodiversity and Conservation, Environmental Pollution, Environmental Policies, Human Population and Swachh Bharat
Publications

Publication List

- Kaushik, M., Pruthi, M., Sharma, A., Sharma, R. S., Mishra, V., Karmakar, S., & Sarkar, N. (2025). Exploring the parallels between nanoparticles and viruses with emphasis on environmental roles and remediation. *Progress in Biophysics and Molecular Biology, 198,* 39–54.

- Karmakar, S., Mukherjee, P., Mishra, V., Gupta, R. K., Kumar, R., Srivastava, P., & Sharma, R. S. (2024). Microhabitat influences on phage-bacteria dynamics in an abandoned mine for ecorestoration. *Journal of Environmental Management, 370*, 1–12.

- Karmakar, S., Saurav, G. K., Tangjang, S., Yadav, K., Yirang, L., & Paron, O. (2024). Phage therapy: a promising cure for bacterial infections in humans. *Journal of Bioresources, 2*, 19–25.

- Mukherjee, P., Sharma, R. S., Rawat, D., Sharma, U., Karmakar, S., Yadav, A., & Mishra, V. (2024). Microbial communities drive flux of acid orange 7 and crystal violet dyes in water-sediment system. *Journal of Environmental Management, 351*, 1–12.

- Kumari, R., Suman, K., Karmakar, S., Mishra, V., Lakra, S. G., Saurav, G. K., & Mahto, B. K. (2023). Regulation and safety measures for nanotechnology-based agri-products. *Frontiers in Genome Editing, 5*, 1–12.

- Sharma, R. S., Karmakar, S., Kumar, P., & Mishra, V. (2019). Application of filamentous phages in environment: A tectonic shift in the science and practice of ecorestoration. *Ecology and Evolution, 9*, 2263–2304.

- Sharma, R. S., Nayak, S., Malhotra, S., Karmakar, S., Sharma, M., Raiping, S., & Mishra, R. (2019). Rhizosphere provides a new paradigm on the prevalence of lysogeny in the environment. *Soil and Tillage Research, 195*, 1–6.

- Rawat, D., Sharma, R. S., Karmakar, S., Arora, L. S., & Mishra, V. (2018). Ecotoxic potential of a presumably non-toxic azo dye. *Ecotoxicology and Environmental Safety, 148,* 528–537.

- Mishra, R., Archana, Sharma, S., Singh, S., Sharma, R. S., Malhotra, S., Karmakar, S., Sardesai, M. M., & Mishra, V. (2017). Purification and characterization of a lectin isoform of ribosome inactivating protein from *Viscum articulatum* parasitic on *Grewia tilifolia*. *Phytomorphology, 67(3&4)*, 91–100.

- Malhotra, S., Mishra, V., Karmakar, S., & Sharma, R. S. (2017). Environmental predictors of indole acetic acid producing rhizobacteria at fly ash dumps: Nature-based solution for sustainable restoration. *Frontiers in Environmental Science, 5*, 1–12.

- Meena, S. S., Sharma, R. S., Gupta, P., Karmakar, S., & Aggarwal, K. K. (2016). Isolation and identification of *Bacillus megaterium* YB3 from an effluent contaminated site efficiently degrades pyrene. *Journal of Basic Microbiology, 55*, 1–10.

- Sharma, R. S., Karmakar, S., & Mishra, V. (2016). Environmental toxicants and reproductive health - an environmental perspective. *ISSRF News Letter, 18,* 78–81.

- Karmakar, S., Mishra, V., & Sharma, R. S. (2016). Environmental applications of ssDNA bacterial viruses. *VirusDisease, 27*, 405–459.

Chapters in books and e-chapters (Published)

- Malhotra, S., Mishra, R., Karmakar, S., Rawat, D., Sharma, S., Singh, S., Kumar, P., Raiping, S., & Mishra, V. (2018). Environmental factors shaping plant-microbe associations: Their significance to improve plant health and vegetation restoration in degraded lands. In *Introduction to Challenges and Strategies to Improve Crop Productivity in Changing Environment* (1st ed., pp. 1–16).

- Karmakar, S., Sharma, R. S., & Mishra, V. (2020). Gandhian philosophy as a driver of sustainable development. In *Gandhi Across the Boundaries* (1st ed., pp. 130–132).

- Karmakar, S. (2020). Environmental Legislations in India. In *Environmental Science Study Material-2* (1st ed.).

- Kumari, R., Saurav, G. K., Suman, K., Lakra, S. G., Karmakar, S., Kumari, P., Islam, A., Yirang, L., & Mahto, B. K. (2025). Developing nutritionally enhanced crops through approaches of functional genomic and CRISPR/Cas technologies. In J.-T. Chen (Ed.), *Stress-Resilient Crops: Coordinated Omics-CRISPR-Nanotechnology Strategies*. CABI Digital Library, Oxfordshire, UK. ISBN: 978-1800-627-284.

Awards and Distinctions

1. CSIR Foreign Travel Grant for USA
2. INSA Foreign Travel Grant for United States of America
3. TOP DOWNLOADED PAPER 2018-2019, Swagata Karmakar whose paper has been recognized as one of the most read in Ecology and Evolution, Wiley Publication.

Association With Professional Bodies

1. Association of Microbiologists of India - Life member
2. Indian Virological Society -Life member
3. International Society for Viruses of Microorganisms - Life member
4. Indian Society for the Study of Reproduction and Fertility – Life member

Signature of Faculty Member